1

Types of Databases

- Relational DB
- Object-Oriented DB
- Hierarchical DB
- N/w DB

Relational DB

- Based on relational data model
- Entities (Real-world objects) → Attributes (features)
- Relations : Tables

Student Relation

Stu-ID	Name	Address	Phone	Course-id

Primary key

Course-Relation → foreign key

Course-id	Teacher	Marks

Object - Oriented DBMS

- Entities + Attributes are stored with entire data set → OBJECT

OOPS
- Encapsulation (show desired data)
- Inheritance
- Polymorphism
- Object Identity

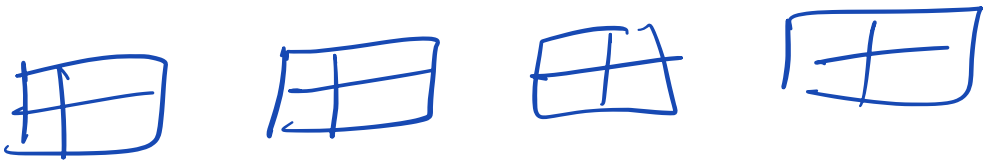
RDBMS
- Transaction control
- ACID
- Integrity
- Consistency

→ Object Oriented Data Model

UNIVERSITY → object

Attributes

Student | Faculty | Salary | Course



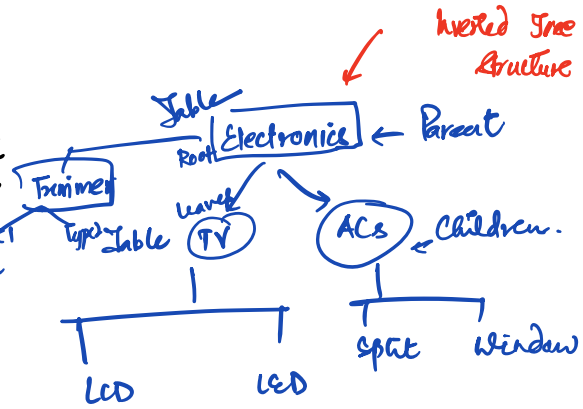
Advantages →

- Complex data saved and retrieved.
- Everything related to OOPs and works perfectly with languages like JAVA
- Object - ID automatically assigned.

Disadvantages

- High complexity of data makes read, write, update, delete slow
- Don't have lot of support

Hierarchical Database



Hierarchical Data Model

(+ves)

- Easy to map any table
  - 1 : 1
  - 1 : N
- Quick retrieval link
- Built-in referential integrity

(-ves)

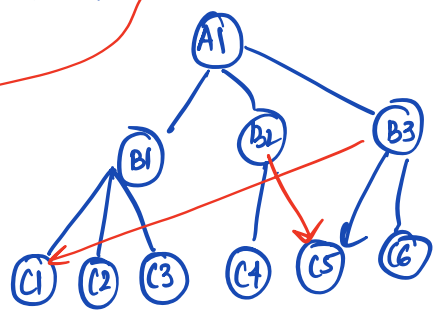
- Child entry has to have a parent entry
- Cannot be used to show complex relations

Network Database

- Modified way of hierarchical database

1 : 1
1 : N

Many to Many



(+) • more complex relationships than hierarchical database

(−) Maintaining n/w database is very tedious

(−) m : N → Slow retrieval

unlike hierarchical 1:1 & 1:N → fast Retrieval